

Forgety

A future-proof platform for forgetting

*Carolina Kelsch
Leonardo Benitez
Natnael Mola*

Context

- Sometimes we need to remove knowledge from a trained model
- Right-to-be-forgotten, copyright infringement, etc
- Many companies use superficial workarounds
- Performing proper unlearning is not a trivial science, with several hidden tradeoffs and poor standardization



AI Firm Midjourney Faces Copyright Lawsuit by Warner Bros., DC Comics and Cartoon Network

Disney and Universal also sued Midjourney for copyright infringement in June.



Katelyn Chedraoui

Sept. 4, 2025 2:46 p.m. PT

3 min read ↗

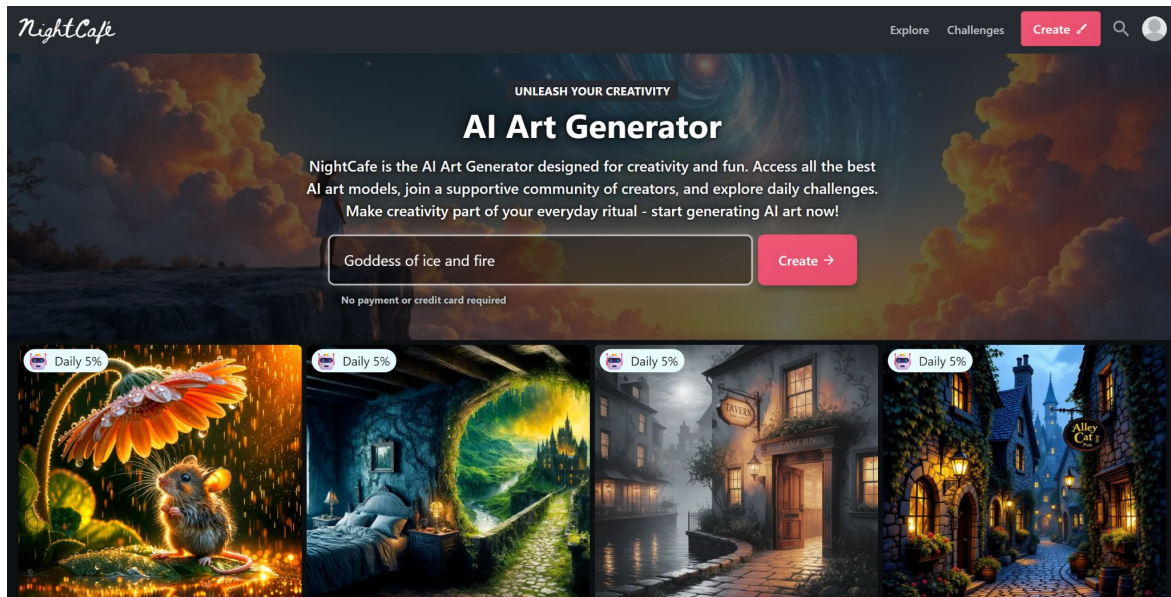
The solution: **Forgety**

- **Unlearning as a Service**
- Consumption-based unlearning platform
- You give model + data, we return the unlearned model to you
- Behind the scenes, we do: algorithm selection, hyperparameter optimization, infrastructure configuration... all the boring part 😊
- Why would someone use Forgety?
 - Delete user data
 - Remove offensive content
 - Comply with copyright
 - Edit knowledge, update brand identity

Use case: Compliance to copyright

An artist, client of a generative ai artistic platform, requests the removal of his work from the model.

Customer/user: NightCafe engineer



MVP

Features

- Submit new requests (zip files, links, model checkpoints, metadata)
- List requests

Main tasks

- Devops/tooling setup
- Backend (receive requests and communication with PPKE cluster)
- Common interface for unlearning methods
- Integrate with Hugging Face (model and metrics)
- Basic UI

Out of scope

- Authentication
- Multi-customer (all requests will belong to the same “customer”)
- Reliable metadata persistence (we’ll save local json files, not in a database)
- Elegant connection with cluster (we’ll run bash commands via SSH)
- Few unlearning algorithms (the simplest)
- Deployment (we’ll run locally)
- Pretty UI

Tech stack

Automation / devops:

Docker, Makefile, Mypi, PEP8, Pytest, Pypi, Readthedocs, Git, Github (issues and task board)

Frontend:

Streamlit, Python

Backend:

FastAPI, Pydantic, JsonSchema, Python

Infrastructure / Cluster integration:

SSH, Slurm commands, Apptainer, Bash

Use case: Compliance to copyright

1. **Customer Request:**

- Provides the model (or HF link), the “forget dataset” (the 200 paintings) and a “retain dataset” of other artworks and general prompts.
- Customer will see pending request.

2. **Unlearning Execution:**

- The model and problem type are evaluated to define the best method to erase the knowledge and inference of unlearning hyperparameters, then the platform runs the unlearning method.
 - Customer can get the status of running unlearning job.
- 

Use case: Compliance to copyright

3. **Validation:**

- Membership inference tests are automatically performed, plus evaluation metrics such as unlearning accuracy, retain accuracy, CLIP score etc.
- Prompt-based evaluations are performed and the results are stored in a report manner.

4. **Delivery:**

- The checkpoint for the unlearned model is uploaded to the customer Hugging Face with a report of the unlearning performance metrics.
 - Customer will see a job finished status and can verify on HF the checkpoints for the unlearned model + unlearn performance metrics.
- 

Development division

All development fronts run in
parallel

Frontend (FE)

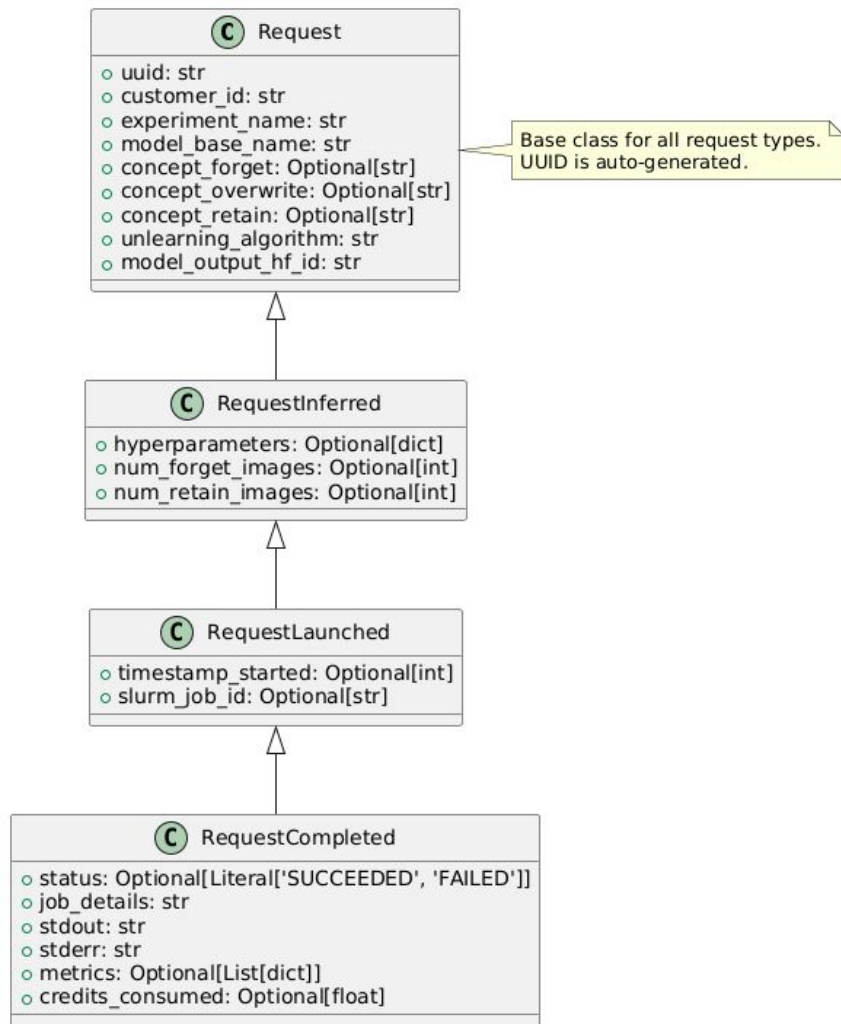
Backend (BE)

ML pipeline (ML)

Testing (T)

Classes - User Request

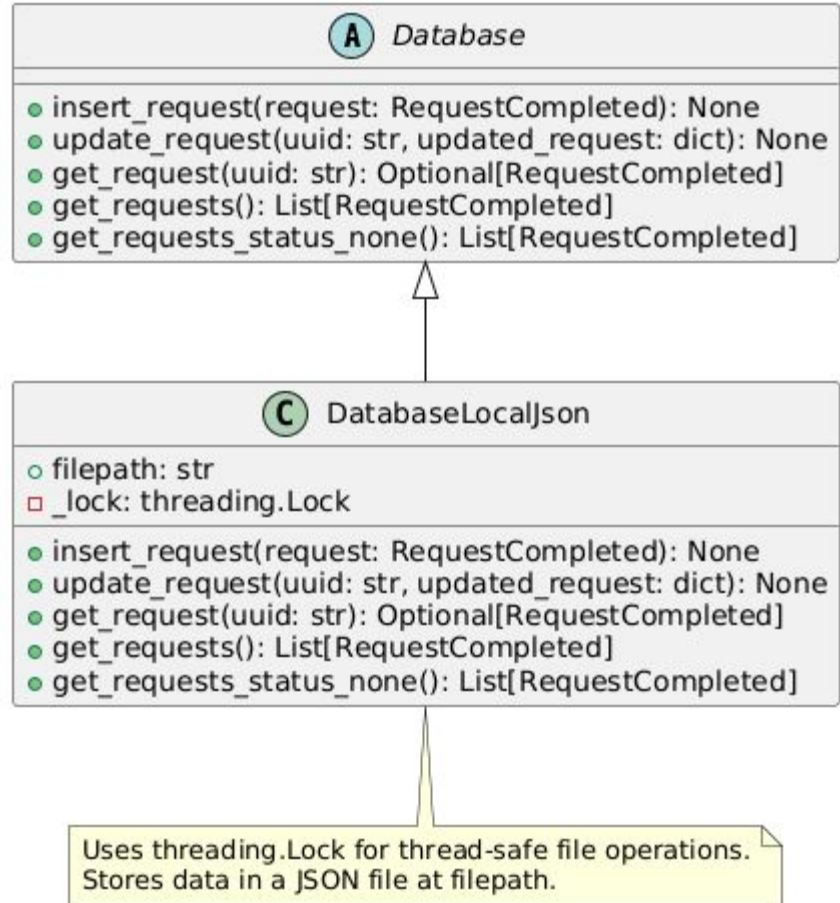
- Handle the request information
- Hold data for backend service
- Keep status of the jobs
- Child classes successively specify as we have more information about the job



Classes - Database

Holds **all information** on the launched jobs by tracking UUID of the unlearning processes.

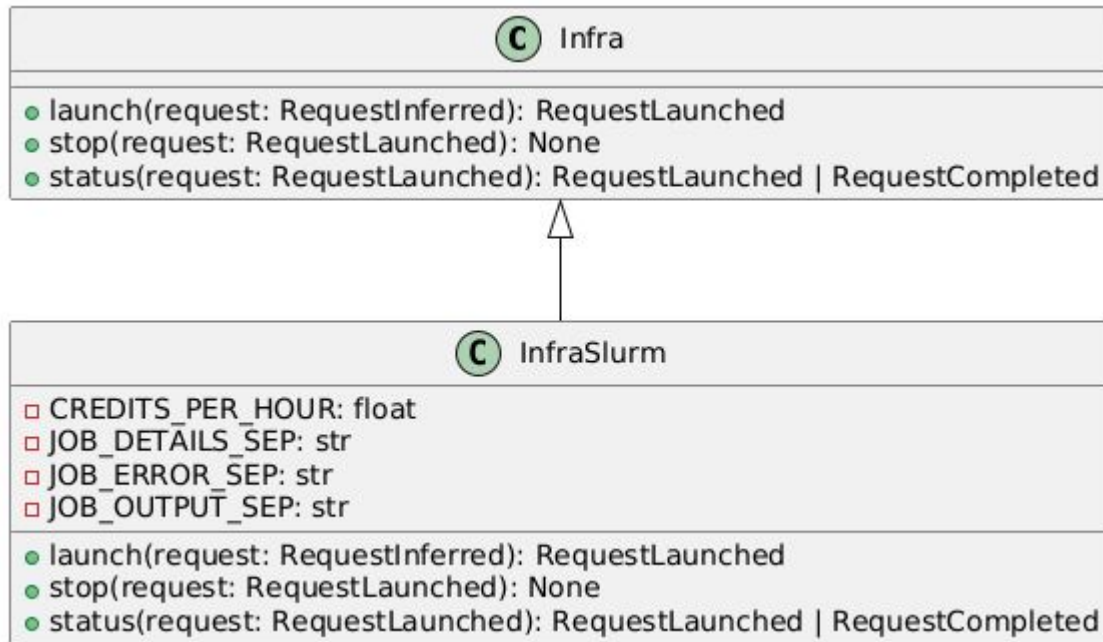
In the future, it can be adapted to other storage structures as SQL, or unstructured DB as Mongo.



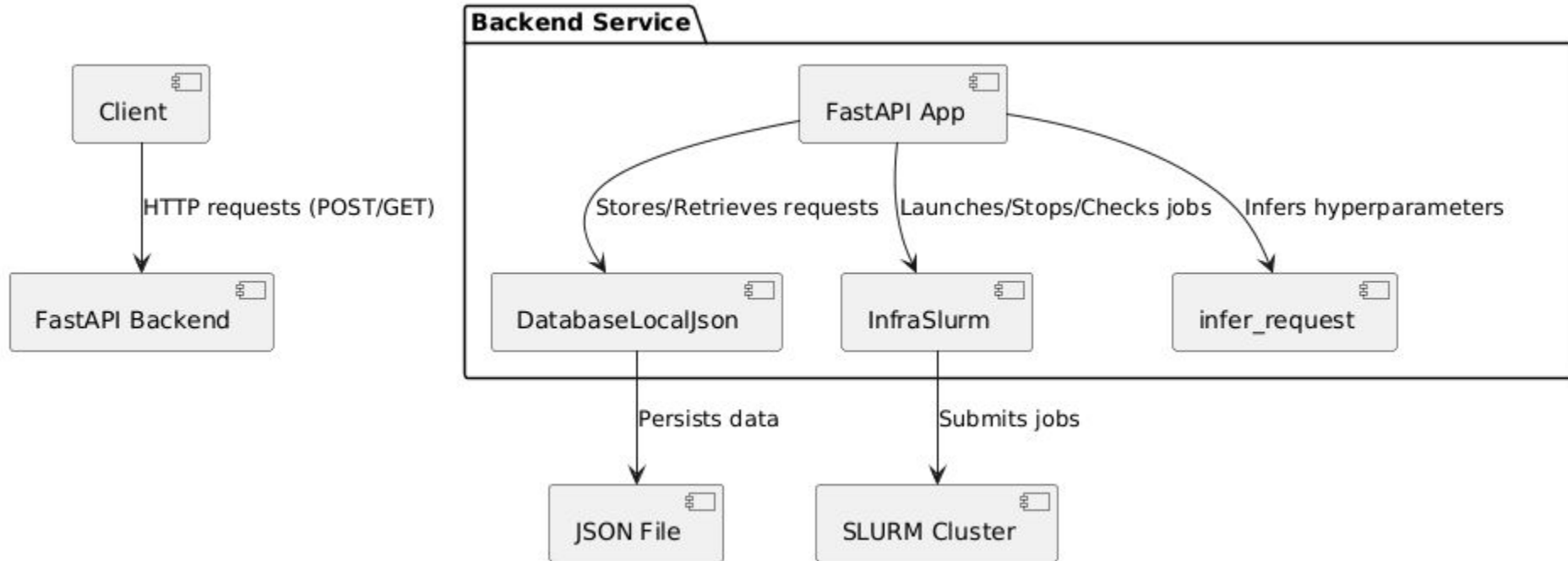
Classes - Infrastructure

Performs the infrastructure communication, serve as an interface for the **computing service**.

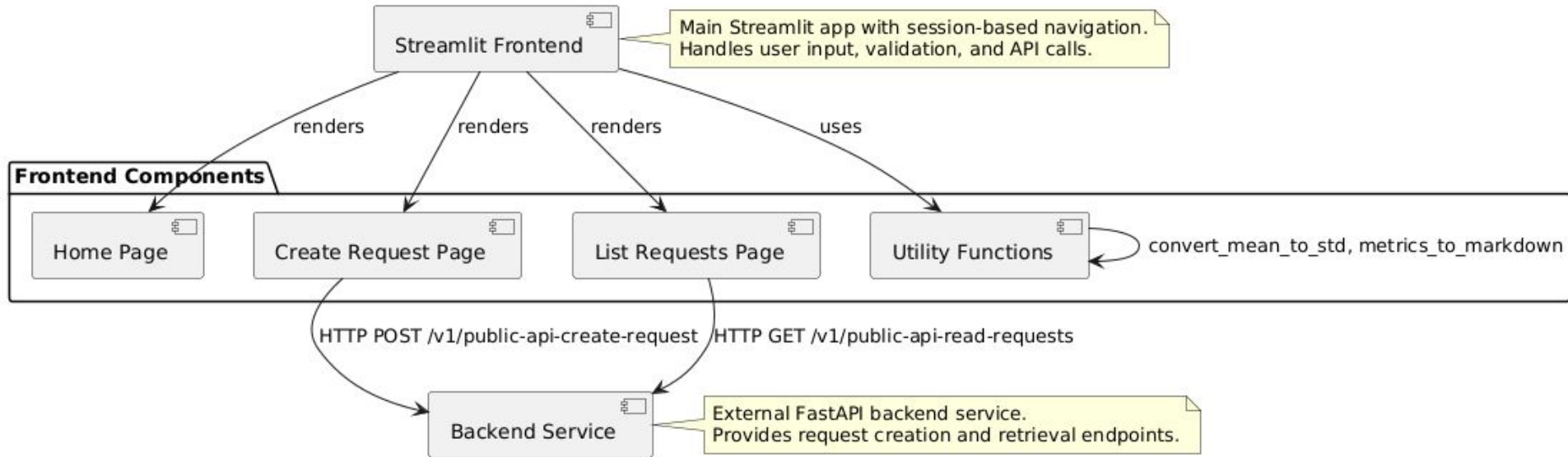
Could also be a K8S cluster, AWS/Azure/other cloud, etc



UML Component Diagram - Backend

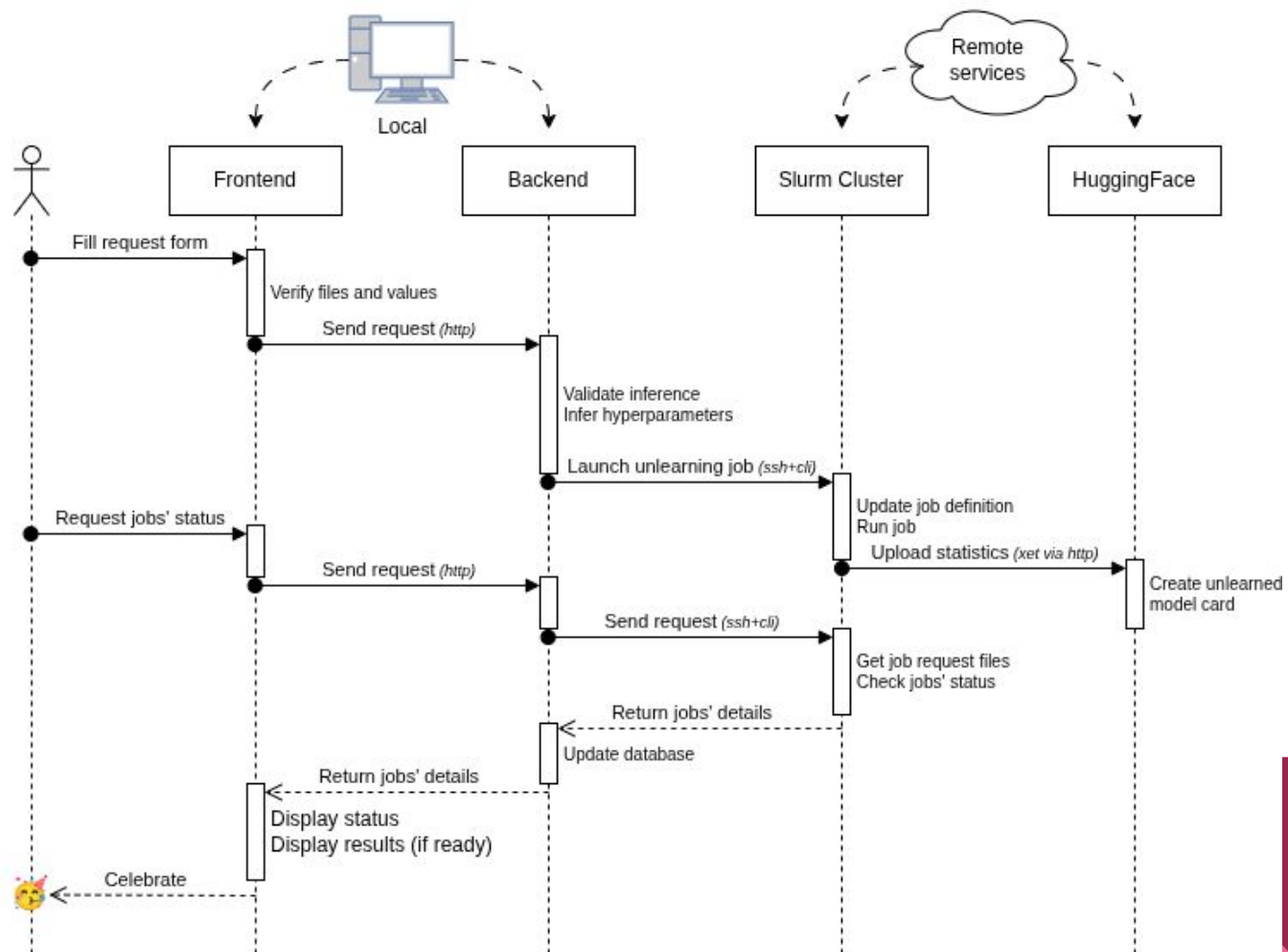


UML Component Diagram - Frontend



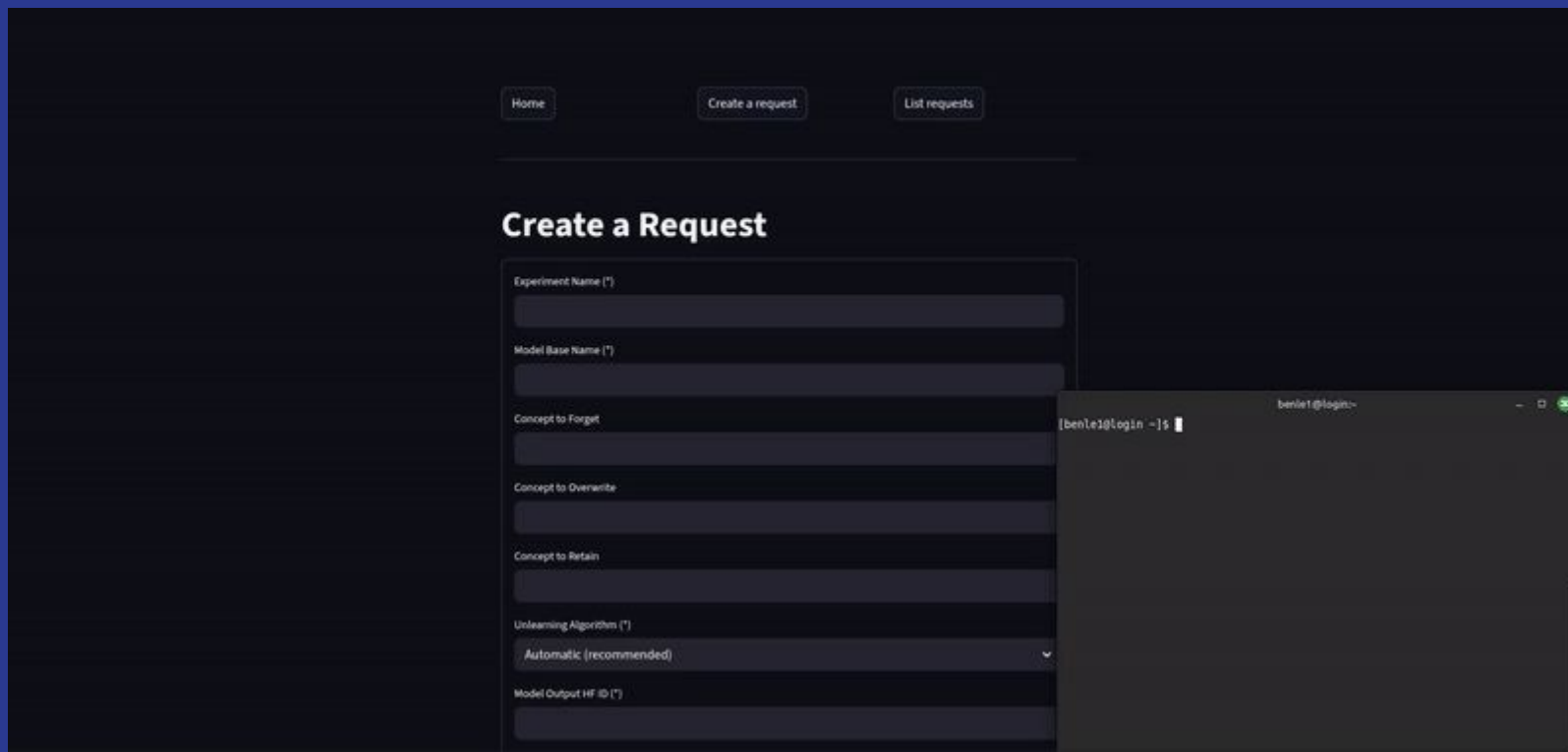
Architecture

UML Sequence Diagram



Demo

<https://youtu.be/mW0lhEiT0Vw>



The screenshot shows a web application interface with a dark theme. At the top, there are three buttons: "Home", "Create a request", and "List requests". Below these buttons is a section titled "Create a Request". This section contains a form with several input fields and a dropdown menu. The form fields are labeled: "Experiment Name (*)", "Model Base Name (*)", "Concept to Forget", "Concept to Overwrite", "Concept to Retain", "Unlearning Algorithm (*)", and "Model Output HF ID (*)". The "Unlearning Algorithm" dropdown menu is currently set to "Automatic (recommended)". To the right of the form, there is a terminal window showing a command prompt with the text "benle1@login:~" and a prompt character "\$".

Home Create a request List requests

Create a Request

Experiment Name (*)

Model Base Name (*)

Concept to Forget

Concept to Overwrite

Concept to Retain

Unlearning Algorithm (*)

Automatic (recommended)

Model Output HF ID (*)

benle1@login:~

```
[benle1@login ~]$
```



Thank you!

Questions & suggestions?